



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO SRI CHELLIAMMAN AGENCY

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

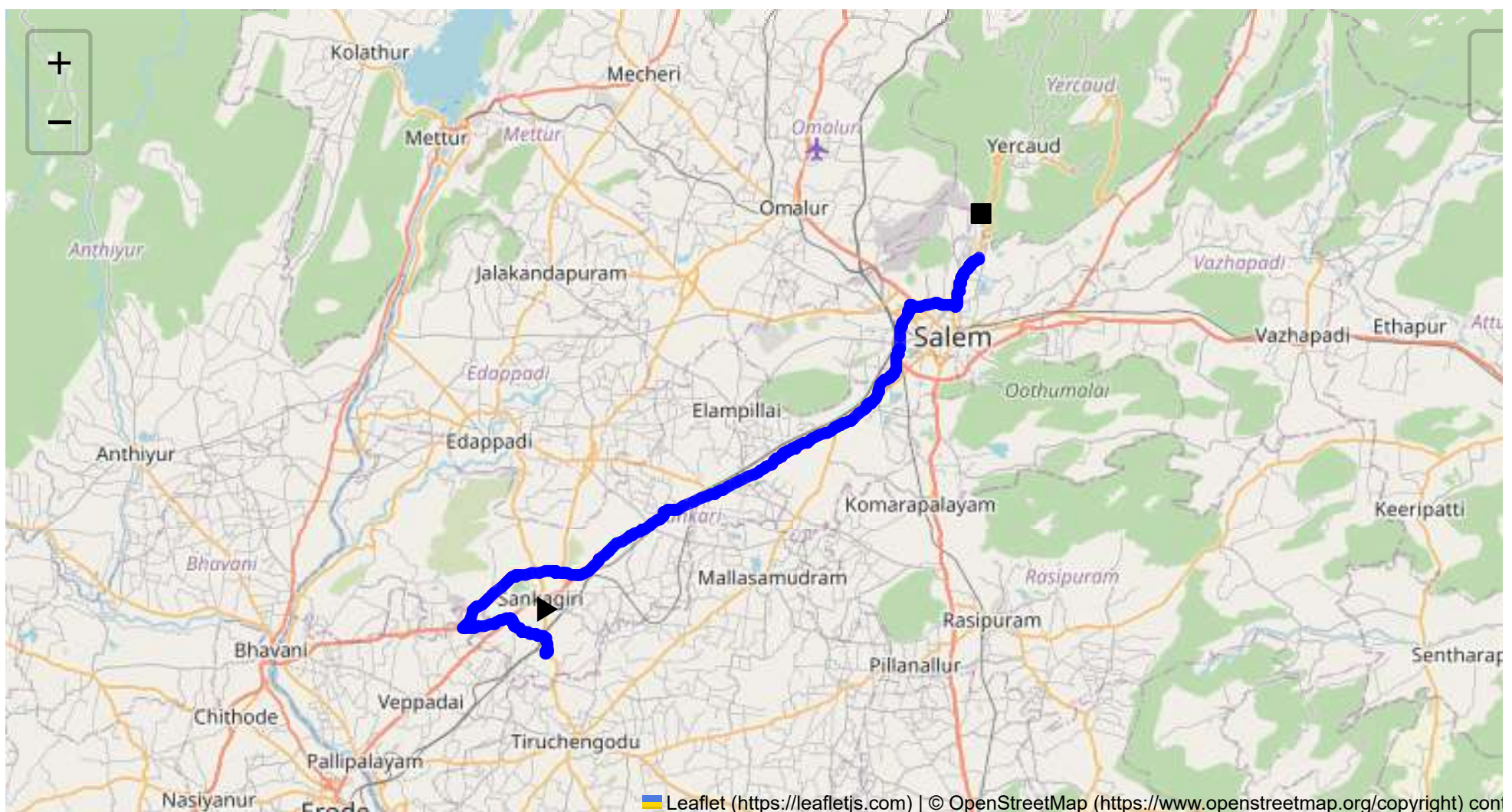
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 61.50 km**  
**Estimated Duration: 1.3 hours**  
**Adjusted Duration (Heavy Vehicle): 1.6 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.708158, 78.176083)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from CVQF+23W in Sangagiri to 164/1A2 in Kondappanaickenpatti, Salem, covers approximately 61.50 kilometers. The journey passes through several key points including Puthur, Padaiveedu, and segments on major highways such as the Salem-Kochi Highway. The route is a mix of rural roads and busy highways, requiring careful navigation through both densely populated areas and more rural settings.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate. Typical weather includes hot summers (March to May), a monsoon season (June to September), and mild winters (October to February). During the monsoon, the route is susceptible to heavy rains which can cause flooding and reduced visibility, increasing the risk of accidents. The summer heat can be intense, possibly leading to road surface damage like asphalt bleeding.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, peak hours are in the morning (7-9 AM) and evening (5-7 PM), when commuters travel to and from work.
- **Congestion-Prone Areas:** Key areas with consistent congestion include urban sections entering Salem, particularly around key city entry points and near major intersections such as the ones in Nedunchalai Nagar.

## 4. Assessment of Road Quality and Infrastructure

The highway sections like the Salem-Kochi Highway are generally well-maintained with proper signage and lighting, but rural roads might be narrow, with potholes and less frequent maintenance. Road quality may vary, impacting driving speed and safety for larger vehicles.

## 5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** In case of road blockages, an alternative could be taking the NH544 and NH544H, which run parallel to some of the sections of the planned route.
- **Bypass Route:** Utilizing the Salem Bypass can help avoid city congestion during peak times.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

In Tamil Nadu, transport of hazardous materials requires strict adherence to regulations such as proper labeling, restricted timings for transportation, and driver licensing requirements. Heavy fines apply for non-compliance. Only certain routes are approved for hazardous materials to ensure public safety.

## 7. Overview of Historical Incidents

Historically, the region has seen incidents involving tanker lorries due to high speed or improper handling during monsoon seasons, most prominently near the highway interchanges where heavy traffic frequency is observed. Such incidents often lead to temporary road closures and traffic diversions.

## 8. Environmental Considerations and Sensitive Areas

Several environmentally sensitive zones exist, including agricultural lands and small water bodies, which could be severely impacted by spills or accidents involving hazardous materials. Local fauna might be

affected as these roads also border rural and semi-forested areas.

## 9. Analysis of Communication Coverage

Communication coverage is generally good along the major highways, but there might be dead zones as you move into rural areas like Vaikundam. Mobile network reception can be patchy, especially during adverse weather conditions.

## 10. Estimated Emergency Response Times

Emergency response in urban areas like Salem can be relatively quick (15-20 minutes), but response times may increase to over 30 minutes in rural sections due to distance and road conditions. Emergency facilities are more readily available in urban centers.

## 11. Overall Summary of Risk Assessment

The route carries standard risks associated with a mix of urban and rural driving, with specific concerns regarding weather impact, road conditions, and traffic congestion during peak hours. Proper planning, adherence to regulations, and real-time monitoring of road conditions can mitigate many potential hazards. Drivers should be particularly vigilant about road quality fluctuations and communication issues in more remote areas.

Overall, the most significant risks involve weather-related impacts, adherence to hazardous material transport regulations, and managing urban congestion effectively. Planning for alternate routes and understanding emergency procedures can help minimize risks during the drive.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.67557, 78.15976 | 15 KM/Hr    |
| 1  | Turn       | Medium     | 11.43822, 77.87348 | 30 KM/Hr    |
| 2  | Turn       | High       | 11.43961, 77.87341 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.44898, 77.87410 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.45357, 77.85789 | 30 KM/Hr    |
| 6  | Turn       | Medium     | 11.45352, 77.85698 | 30 KM/Hr    |
| 7  | Turn       | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.46048, 77.85036 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.46303, 77.84945 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.46314, 77.84928 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 11 | Turn      | High       | 11.45542, 77.81725 | 15 KM/Hr    |
| 12 | Turn      | High       | 11.45631, 77.81668 | 15 KM/Hr    |
| 13 | Turn      | High       | 11.49409, 77.88004 | 15 KM/Hr    |
| 14 | Turn      | Medium     | 11.49419, 77.88015 | 30 KM/Hr    |
| 15 | Turn      | Medium     | 11.49365, 77.88596 | 30 KM/Hr    |
| 16 | Turn      | High       | 11.49236, 77.89552 | 15 KM/Hr    |
| 17 | Turn      | High       | 11.49225, 77.89560 | 15 KM/Hr    |
| 18 | Turn      | Medium     | 11.60942, 78.10056 | 30 KM/Hr    |
| 19 | Turn      | Medium     | 11.60965, 78.10070 | 30 KM/Hr    |
| 20 | Turn      | Medium     | 11.61736, 78.10588 | 30 KM/Hr    |
| 21 | Turn      | Medium     | 11.61745, 78.10601 | 30 KM/Hr    |
| 22 | Turn      | Medium     | 11.64310, 78.11982 | 30 KM/Hr    |
| 23 | Turn      | Medium     | 11.64490, 78.12024 | 30 KM/Hr    |
| 24 | Turn      | Medium     | 11.64503, 78.12034 | 30 KM/Hr    |
| 25 | Turn      | High       | 11.67074, 78.12668 | 15 KM/Hr    |
| 26 | Turn      | High       | 11.67081, 78.12661 | 15 KM/Hr    |
| 27 | Turn      | High       | 11.67582, 78.12786 | 15 KM/Hr    |
| 28 | Turn      | Medium     | 11.69518, 78.16364 | 30 KM/Hr    |
| 29 | Turn      | Medium     | 11.70719, 78.17515 | 30 KM/Hr    |

## Emergency Locations

|   | type     | name                                                            | coordinates               | speed_limit | risk_level |
|---|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 2 | hospital | C. N. Hospital                                                  | 11.55724, 78.0096         | 30 km/h     | Medium     |
| 3 | hospital | Annapoorna Medical College Hospital                             | 11.57693,<br>78.037627    | 30 km/h     | Medium     |
| 4 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 5 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 6 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |

|    | type     | name                                                   | coordinates               | speed_limit | risk_level |
|----|----------|--------------------------------------------------------|---------------------------|-------------|------------|
| 9  | hospital | Vinayaka Mission's Kirupananda Variyar Medical College | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 13 | hospital | Bharathi Hospital, Salem                               | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 14 | hospital | Aishwarya Hospital                                     | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 15 | hospital | C.M. Hospital                                          | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |
| 16 | pharmacy | Healthy Pharmacy                                       | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |
| 17 | pharmacy | Thulasi Pharmacy                                       | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 19 | hospital | Venkatesh Hospital and Urology Centre                  | 11.6759517,<br>78.1324029 | 30 km/h     | Medium     |
| 20 | hospital | R.K. Hospital, Salem                                   | 11.6746905,<br>78.1339717 | 30 km/h     | Medium     |
| 21 | hospital | Kurinji Hospital, Salem                                | 11.6771806,<br>78.1351224 | 30 km/h     | Medium     |
| 22 | clinic   | Ayush Care Siddha Centre                               | 11.6731195,<br>78.1351967 | 30 km/h     | Medium     |
| 23 | clinic   | Bismillah Piles Clinic                                 | 11.6755285,<br>78.1343325 | 30 km/h     | Medium     |
| 24 | hospital | Bhavani Hospital                                       | 11.6745697,<br>78.1362683 | 30 km/h     | Medium     |
| 25 | clinic   | Arunagiri Ent Nursing Home                             | 11.6762823,<br>78.1369593 | 30 km/h     | Medium     |
| 26 | clinic   | Srimaya Child Health Centre                            | 11.6750852,<br>78.1370646 | 30 km/h     | Medium     |
| 27 | hospital | Gl Hospital                                            | 11.6742597,<br>78.1379092 | 30 km/h     | Medium     |
| 28 | clinic   | Ayush Multispeciality Homoeopathic Clinic              | 11.6770982,<br>78.1395947 | 30 km/h     | Medium     |
| 29 | hospital | Balaji Skin Hospital                                   | 11.676166,<br>78.1379488  | 30 km/h     | Medium     |
| 30 | hospital | Dr. Sasi Eye Care                                      | 11.6743698,<br>78.1379234 | 30 km/h     | Medium     |
| 31 | police   | Alagapuram Police station                              | 11.6769953,<br>78.1422615 | 30 km/h     | Medium     |
| 32 | hospital | AP medical center                                      | 11.6774074,<br>78.1425988 | 30 km/h     | Medium     |

|    | type     | name                                         | coordinates               | speed_limit | risk_level |
|----|----------|----------------------------------------------|---------------------------|-------------|------------|
| 33 | hospital | M G Diabetes Speciality and Research Centre  | 11.6766146,<br>78.1422986 | 30 km/h     | Medium     |
| 34 | hospital | Nathan Super Speciality Hospital             | 11.6743874,<br>78.1417222 | 30 km/h     | Medium     |
| 35 | hospital | Devi Hospital                                | 11.6786391,<br>78.1415731 | 30 km/h     | Medium     |
| 36 | hospital | Fairlands Eye Cure                           | 11.677959,<br>78.1415897  | 30 km/h     | Medium     |
| 38 | hospital | Dr. Soodamani Hospital                       | 11.6757031,<br>78.14473   | 30 km/h     | Medium     |
| 39 | hospital | Kavin Eniyan Hospital                        | 11.677643,<br>78.144129   | 30 km/h     | Medium     |
| 40 | hospital | Sri Vaari Hospital                           | 11.6763726,<br>78.1467277 | 30 km/h     | Medium     |
| 41 | hospital | Krishna Speciality Hospital                  | 11.676751,<br>78.145561   | 30 km/h     | Medium     |
| 44 | hospital | Gopi Memorial Hospital                       | 11.673855,<br>78.1516078  | 30 km/h     | Medium     |
| 45 | clinic   | Saraswathi Nursing Home                      | 11.6746327,<br>78.1520981 | 30 km/h     | Medium     |
| 46 | hospital | J S Hospital, Dharmapuri                     | 11.6739064,<br>78.1521557 | 30 km/h     | Medium     |
| 47 | hospital | Salem Gopi Hospitals                         | 11.6745894,<br>78.1515482 | 30 km/h     | Medium     |
| 48 | hospital | Dr. P. Sundarrajan Neuro Hospital            | 11.6742791,<br>78.1499738 | 30 km/h     | Medium     |
| 49 | hospital | Mahadevas Ayurvedic Hospital                 | 11.6752045,<br>78.1517396 | 30 km/h     | Medium     |
| 50 | clinic   | Akshaya Fertility Centre                     | 11.6739088,<br>78.1518657 | 30 km/h     | Medium     |
| 52 | hospital | Shanmuga Hospital and Salem Cancer Institute | 11.6770178,<br>78.1531844 | 30 km/h     | Medium     |
| 53 | hospital | Tms Eye Hospital                             | 11.6772791,<br>78.1536202 | 30 km/h     | Medium     |
| 54 | hospital | Sureka Hospital                              | 11.676423,<br>78.1550511  | 30 km/h     | Medium     |
| 56 | hospital | Ramana Surgical Hospital                     | 11.6731205,<br>78.15862   | 30 km/h     | Medium     |
| 57 | hospital | Salem Medical Centre                         | 11.67607,<br>78.163014    | 30 km/h     | Medium     |



|    | type     | name                    | coordinates               | speed_limit | risk_level |
|----|----------|-------------------------|---------------------------|-------------|------------|
| 58 | hospital | KS Siddha Hospital      | 11.6752685,<br>78.1633063 | 30 km/h     | Medium     |
| 59 | clinic   | Sindhu Eye Care Centre  | 11.6814684,<br>78.1625895 | 30 km/h     | Medium     |
| 61 | hospital | Kiran Diabetic Hospital | 11.6957337,<br>78.1625792 | 30 km/h     | Medium     |
| 62 | hospital | Malar Hospital          | 11.7041061,<br>78.16998   | 30 km/h     | Medium     |

### Crowded Spots

|    | type       | name                                                       | coordinates               | speed_limit | risk_level |
|----|------------|------------------------------------------------------------|---------------------------|-------------|------------|
| 0  | school     | KRP Matric. Hr. Sec School                                 | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 7  | university | Vinayaka Missions Vinayaka Missions<br>Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 8  | school     | Vinayaka Missions Annapoorani Matric<br>School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 10 | school     | Dist Education Training                                    | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 11 | school     | DIET - Salem                                               | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 12 | school     | Diamond Rays Matric HR Sec School                          | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |
| 18 | college    | Sona College                                               | 11.6768,<br>78.1257052    | 30 km/h     | Medium     |
| 37 | school     | Saradha Balamandir School                                  | 11.6771182,<br>78.1472461 | 30 km/h     | Medium     |
| 42 | school     | Saradha College                                            | 11.6769525,<br>78.1481901 | 30 km/h     | Medium     |
| 43 | school     | Sri Sarada Matriculation Higher<br>Secondary School        | 11.6768969,<br>78.1488115 | 30 km/h     | Medium     |
| 51 | school     | Holy flower matric hr.sec school                           | 11.6748637,<br>78.1528099 | 30 km/h     | Medium     |
| 55 | school     | CSI Hobart Edu Institute                                   | 11.673245,<br>78.1567934  | 30 km/h     | Medium     |

|    | type    | name     | coordinates               | speed_limit | risk_level |
|----|---------|----------|---------------------------|-------------|------------|
| 60 | college | Govt ITI | 11.6919383,<br>78.1636731 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.67557, 78.15976



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.44898, 77.87410



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45357, 77.85789





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46048, 77.85036





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46314, 77.84928





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45542, 77.81725



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45631, 77.81668



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49409, 77.88004





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49419, 77.88015

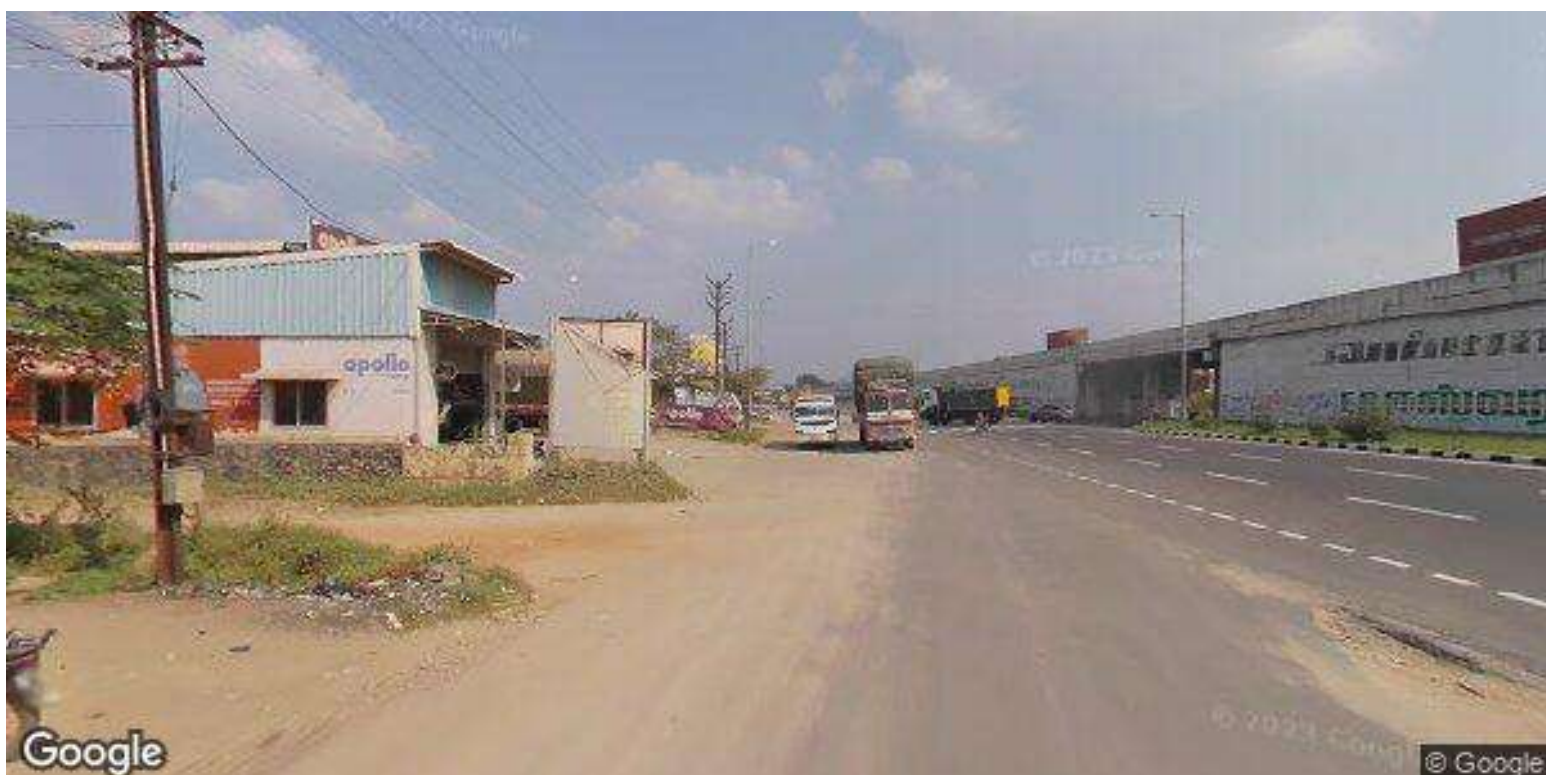


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49365, 77.88596





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49236, 77.89552



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49225, 77.89560



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.60942, 78.10056





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.60965, 78.10070



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.61736, 78.10588





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.61745, 78.10601**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.64310, 78.11982**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.64490, 78.12024**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64503, 78.12034



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.67074, 78.12668





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.67081, 78.12661



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.67582, 78.12786



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.69518, 78.16364





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.70719, 78.17515

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